

Exact weight maxima at $k \leq 3$: $W_{\max} = 5, 6, 7$ and proven pool radii

Proof document (appendix draft, oracle-independent), companion to the $k \geq 4$
weight-ceiling document

A. Longo Claude (CC)*

July 10, 2026

Abstract

For edge-to-edge tilings of the plane by unit-edge regular polygons (3, 4, 6, 12-gons, plus the analytically solved 4.8.8 octagon tiling), we prove exact values and proven search radii for $W(T)$ — the least possible maximum step-weight of a basis of the period lattice Λ — in the regime $k \leq 3$ that the *pgg* ceiling document does not cover. Main results: (i) branch-wise upper bounds, by the shape of Λ , each either a short lemma or a certified finite computation in exact arithmetic; (ii) exactness: $\max W = 5, 6, 7$ at $k = 1, 2, 3$, attained by *p6m* hexagonal lattices, with length-based lower-bound certificates; (iii) a consumption analysis showing the existing tuned $k = 3$ enumeration pool is *not* covered by these bounds (both its weight radius and its Euclidean caps clip proven shells), so the certified $k = 3$ run must be re-executed at the proven configuration. The key device is that for rigid (hexagonal or square) lattices the minimal vector’s exact norm is pinned by the cell area, the admissible areas form a finite nonnegative census, and each exact norm shell of $\mathbb{Z}[\zeta_{12}]$ is finite — so the density of the ring, fatal to length-based arguments, is irrelevant. No enumeration data is used in any proof; the oracle appears only as a posteriori confirmation.

1 Setting and statement

Let T be a k -uniform edge-to-edge tiling of $\mathbb{R}^2$ by unit-edge regular polygons of the family $\{3, 4, 6, 12\}$, $k \leq 3$, with full symmetry group G , point group P , and period lattice Λ . The octagon is excluded from the family: an octagon forces the unique 4.8.8 tiling, which is 1-uniform (settled decision; its lattice is handled analytically in Theorem 1.1). Consequently all edge directions lie on one 30° grid and $\Lambda \subset \mathbb{Z}[\zeta_{12}]$; weights are step-counts in the 12-direction alphabet (24 directions only for 4.8.8). $W(\Lambda)$ is the minimum over bases of Λ of the maximum weight of a basis vector; a search pool of weight radius s captures Λ by *pairs* iff $W(\Lambda) \leq s$, and by *generators + joins* iff Λ is generated by vectors of weight $\leq s$.

Theorem 1.1 (small- k radii, per branch). *Every period lattice Λ of a k -uniform tiling, $k \leq 3$, satisfies the bound of its row:*

*Architecture, lemmas, computation, and drafting: CC, at A. Longo’s direction (the completeness claim for the $k = 3$ enumeration is the motivating deliverable). Hardened by three adversarial referee subagents; Section 8 documents the process. Source of truth: `docs/SMALLK_W_BOUND.md`, v2, 2026-07-10.

branch (shape of Λ)	$k = 1$	$k = 2$	$k = 3$	form consumed by the solver
hexagonal	6	8	10	basis $(u, \omega u)$: pairs suffice
square	3	6	7	basis (u, iu) : pairs suffice
$\text{hol}(\Lambda) \leq 4$	7	15	23	generators in $W(2n-1)$ + join closure
4.8.8 ($k = 1, \zeta_{24}$)	3	—	—	$u = 1 + \sqrt{2}$, basis (u, iu) , analytic

Theorem 1.2 (exactness). $\max_{T, k\text{-uniform}} W(\Lambda(T)) = 5, 6, 7$ at $k = 1, 2, 3$, attained by $p6m$ tilings with hexagonal lattices (4.6.12 at $k = 1$); certificates in Section 6.

These replace, at $k \leq 3$, the only previously proven radius $24k - 1$ (**thm:weight**), whose $k = 1$ pool was measured past $1.5 \cdot 10^8$ candidate values (budget abort). Pool sizes here are tame: $|W(10)| = 18,481$ and $|W(23)| = 458,713$ in 12 directions, before norm filters.

2 Inputs cited from the thesis

Proven there, used as black boxes here: **lem:wallpaper** (k -uniform tilings have a wallpaper group, in particular a rank-2 period lattice; and distinct vertices are at distance ≥ 1 , so $\lambda_1 \geq 1$); **lem:quotient** ($|V(Q)| \leq k|P|$, where Q is the graph of vertex classes mod Λ); **lem:voltage** and **thm:weight** (Λ is generated by fundamental-cycle voltages of Q , each of weight $\leq 2|V(Q)| - 1$, each edge contributing one unit root); the bound $|P| \leq \text{hol}(\Lambda)$ (inside **prop:orbitfloor**: P is the group of linear parts, injective since Λ carries all translations of G , and preserves Λ since $\Lambda \trianglelefteq G$, so P embeds in the holohedry as a subgroup — whence, by Lagrange, the stronger “ $|P|$ divides $\text{hol}(\Lambda)$ ” used below). The holohedry values $\text{hol} \in \{2, 4, 8, 12\}$ (oblique; rectangular/rhombic; square; hexagonal) are the classical Bravais classification; degenerate rhombic lattices at $60^\circ/90^\circ$ have holohedry $12/8$ and belong to the hexagonal/square branches. Finally the corner-share identity: $|\det \Lambda| = A = \sum_{v \in V(Q)} \text{share}(\text{VC}(v))$, where a corner of a regular m -gon contributes area/ m . In coordinates $(p, q) \triangleq (p + q\sqrt{3})/12$ the shares are: triangle $(0, 1)$, square $(3, 0)$, hexagon $(0, 3)$, 12-gon $(6, 3)$ — all nonnegative.

3 The census lemmas

Lemma 3.1 (orbit census). *Each of the $\leq k$ vertex orbits under G contributes $m_i = |P|/|\text{Stab}_i|$ translation classes, with m_i dividing $|P|$, all of one VC species. Hence $A = \sum_{i \leq k} m_i \text{share}(\text{VC}_i)$ with $|P|$ dividing $\text{hol}(\Lambda)$; and on the quotient torus $V - E + F = 0$ with $E = \frac{1}{2} \sum m_i \deg_i$ and $F = \sum_p t_p$, $t_p = (\text{total } p\text{-corner count})/p \in \mathbb{Z}_{\geq 0}$. The species are the 10 multisets over $\{3, 4, 6, 12\}$ with interior angles summing to 2π .*

Proof. $\text{Stab}_G(v)$ is finite, meets Λ trivially (nontrivial translations are fixed-point-free), and injects into P ; since $\Lambda \trianglelefteq G$, the Λ -classes inside the orbit Gv biject with $P/\overline{\text{Stab}}(v)$, giving $m_i = |P|/|\text{Stab}_i|$, a divisor of $|P|$ by Lagrange. Distinct G -orbits have disjoint class sets, so $n := |V(Q)| = \sum m_i$. The quotient is a genuine CW torus — a nonzero translation cannot fix a bounded tile (interior overlap) nor reverse an edge ($2v = 0$) — so the handshake and Euler identities apply verbatim, with loops and multi-edges handled by the counting form. (Given angle-validity and $t_p \in \mathbb{Z}$, Euler holds identically; the binding filters are the parity and per- p integrality conditions — Euler is kept as belt-and-braces.) The species list is the classical around-a-point enumeration restricted to $\{3, 4, 6, 12\}$, re-derived mechanically in the artifact. $\square$

Lemma 3.2 (shell finiteness). *For $x = a + b\zeta_{12} + c\zeta_{12}^2 + d\zeta_{12}^3 \in \mathbb{Z}[\zeta_{12}]$, $|x|^2 = X + Y\sqrt{3}$ with $X = a^2 + b^2 + c^2 + d^2 + ac + bd \in \mathbb{Z}$ and $Y = ab + bc + cd \in \mathbb{Z}$. The identity $2X - (a^2 + b^2 + c^2 + d^2) = (a+c)^2 + (b+d)^2 \geq 0$ gives $X \geq \frac{1}{2}(a^2 + b^2 + c^2 + d^2)$, so each exact shell $\{x : |x|^2 = X + Y\sqrt{3}\}$ is finite, with coefficients bounded by $\sqrt{2X}$.*

Density of $\mathbb{Z}[\zeta_{12}]$ in $\mathbb{C}$ — which defeats every Euclidean-radius argument — is irrelevant once the norm is pinned exactly. (Referee cross-check: the computed weights agree with the independently derived closed form $\text{wt} = f(a_0, a_2) + f(a_1, a_3)$, $f(p, q) = \min_s |p+s| + |q-s| + |s|$, on all 4,140 shell elements.)

Lemma 3.3 (pure-species rigidity). *If every vertex of T has species $s \in \{3^6, 4^4, 6^3, 3.12.12, 4.6.12\}$, then T is the corresponding uniform tiling; in particular $k = 1$, with (n, shell) : $3^6 : (1, (1, 0)_{\text{hex}})$; $4^4 : (1, (1, 0)_{\text{sq}})$; $6^3 : (2, (3, 0)_{\text{hex}})$; $3.12.12 : (6, (7, 4)_{\text{hex}})$; $4.6.12 : (12, (12, 6)_{\text{hex}})$.*

Proof. $3^6/4^4/6^3$: single tile type; a regular tile placed on a given full edge, on a given side, is unique, so the tiling is forced from any one tile. $3.12.12$: the vertex word admits adjacent pairs $(3, 12)$ and $(12, 12)$ only, so no two triangles share an edge (every neighbour of a triangle is a 12-gon), and at each vertex of a 12-gon its two edges neighbour one 3 and one 12; hence edge-neighbours around every 12-gon alternate 3, 12 (12 edges, even — consistent; the alternation phase of a 12-gon is fixed by any single one of its neighbours). $4.6.12$: pairs $(4, 6)$, $(6, 12)$, $(4, 12)$ only, no same-type pair; neighbours alternate 6, 12 around each square, 4, 12 around each hexagon, 4, 6 around each 12-gon (all even edge counts; phase again fixed by one neighbour). *Propagation (both cases)*: T contains a 12-gon (every vertex has one); the uniform tiling U is transitive on its 12-gons, so position U by an isometry matching one 12-gon of T . Induct on distance r in the tile-adjacency graph from that tile: a tile at distance $r+1$ shares a full edge with a matched tile t at distance r ; its *type* is forced (by t 's alternation pattern, whose phase was fixed when t was matched, or trivially when all of t 's neighbours have one type), and a regular tile of known type on a known full edge and side is uniquely placed — so it matches U . The tile-adjacency graph is connected (Lemma 3.4), so $T = U$; U is vertex-transitive, so $k = 1$. The tabulated (n, shell) values are computed from U 's known structure (e.g. $3.12.12$: $n = 6$, $|u| = 2 + \sqrt{3}$, $|u|^2 = 7 + 4\sqrt{3}$). $\square$

Lemma 3.4 (co-occurrence connectivity). *Let S be the set of species present in T and $C(S)$ the graph on tile types with $t \sim t'$ iff some $s \in S$ contains both. Then $C(S)$, restricted to the types present, is connected. Census combinations with disconnected $C(S)$ are unrealizable.*

Proof. Two tiles sharing an edge are both incident to both of its endpoints, so their types co-occur in the endpoint VCs and are C -adjacent. For *edge-to-edge, locally finite* tilings by disks, the full-edge tile-adjacency graph is connected: perturb a path between two tile interiors off the (discrete) vertex set; each transversal edge crossing switches between the two tiles sharing that full edge. (The hypotheses matter — the claim fails for non-edge-to-edge tilings such as running-bond brick.) Both hold here. Connectivity of the tile graph then forces connectivity of the type image. $\square$

Lemma 3.5 (rigid basis form and pinned shells). *Let Λ be hexagonal-shaped and u a minimal vector. Then $\Lambda = \langle u, \omega u \rangle$ with $\omega = e^{i\pi/3} = \zeta_{12}^2$, $\text{wt}(\omega u) = \text{wt}(u)$, and $|u|^2 = \frac{2}{\sqrt{3}}A$. In census coordinates $A = (P_A + Q_A\sqrt{3})/12$ this gives $(X, Y) = (Q_A/6, P_A/18)$, nonnegative; combinations for which these fail to be integers are unrealizable as hexagonal lattices (by Lemma 3.2 a realizable $|u|^2$ has integer (X, Y)) and are soundly dropped, remaining covered by the other branches' pools, which are consumed as a union. Squares identically with $i = \zeta_{12}^3$, $|u|^2 = A$, $(X, Y) = (P_A/12, Q_A/12)$; the 4.8.8 case with $i = \zeta_{24}^6$.*

Proof. $\text{hol}(\Lambda)$ contains the 60° rotation, so $\omega u \in \Lambda$ and $\langle u, \omega u \rangle \subseteq \Lambda$. Equality: reduce any $x \in \Lambda$ modulo $\langle u, \omega u \rangle$; the remainder lies within the covering radius $|u|/\sqrt{3} < |u| = \lambda_1$ of the triangular lattice $\langle u, \omega u \rangle$ and is itself in Λ , hence is 0. (Square case: covering radius $|u|/\sqrt{2} < |u|$.) Multiplication by a unit root permutes the step alphabet, so $\text{wt}(\omega u) = \text{wt}(u)$, and $W(\Lambda) \leq \text{wt}(u)$. The norm pin is $A = |\det \Lambda| = \frac{\sqrt{3}}{2}|u|^2$ (hex) resp. $|u|^2$ (square); the coordinate arithmetic is immediate. $\square$

4 The computation (rigid branches)

Artifact: `experiments/results/smallk-census-shells-2026-07-10.py` (+ output CSV), exact integer arithmetic throughout, $\sqrt{3}$ carried as coordinate pairs. Pipeline: enumerate census combinations per Lemma 3.1 ($|P|$ over divisors of 12 resp. 8; $r \leq k$ orbits; $m_i \mid |P|$; parity, per- p integrality, Euler), drop pure-rigid multi-orbit combinations (Lemma 3.3) and C -disconnected ones (Lemma 3.4), add the five true rigid shells; map to shells per Lemma 3.5; enumerate every shell element (Lemma 3.2 coefficient bound, boundary asserted); compute wt exactly by breadth-first search over the 12 unit directions (all shells resolved by depth 10); report the per-shell *maximum* (sound: the tiling's u is *some* element of its shell; all six minimal vectors share one weight, but distinct rotation classes on one shell may differ — the maximum covers all).

max shell weight	$k \leq 1$	$k \leq 2$	$k \leq 3$
hexagonal	6	8	10
square	3	6	7

Sanity pins: the 4.6.12 shell (12, 6) has weight exactly 5, its measured W . The $k=1$ hexagonal maximum 6 is attained only by shell (14, 0), whose sole census provenance (re-derived by hand: at $r=1$ the only $m \cdot q = 84$ with $m \mid 12$ is $m = 12$, $q = 7$) is a pure- $3^4.6$ orbit with $m = 12$; the real snub tiling is $p6$ with $m = 6$ — killing this needs a snub-uniqueness lemma, deliberately not claimed, hence 6 rather than 5. The square column carries the same kind of slack (its $k=1$ value 3 comes from a phantom $m=8$ pure- $3^3.4^2$ combination; the real maxima are 1 and 2).

5 The non-rigid branch ($\text{hol} \leq 4$)

$|P| \leq \text{hol}(\Lambda) \leq 4$ gives $n = |V(Q)| \leq 4k$, so `thm:weight` yields generators of weight $\leq 2n - 1 = 7, 15, 23$ at $k = 1, 2, 3$, and Lemma 3.1 bounds the cell area by $A \leq 4k s_{\max}$ with $s_{\max} = (12 + 7\sqrt{3})/12 \approx 2.01$ the maximal VC share ($A \lesssim 24.1$ at $k = 3$; contrast `cor:box(ii)`'s $24k a_{\max} \approx 806$, where $a_{\max} = 3(2 + \sqrt{3})$ is the 12-gon *area* — distinct constants). The join-closure procedure of `cor:box(iv)` then runs over $W(2n-1)$ filtered by $|\det| \leq 24.1$ and the Gauss bounds. This branch is proven with zero new mathematics but is the loose one (the measured non-rigid maximum at $k \leq 3$ is 6); the named tightening is the quotient-first sweep over $|V(Q)| \leq 12$ (work item ST-6), which computes W per voltage lattice directly and bypasses the pool entirely.

6 Attainment: exactness of 5, 6, 7

Lower bound: every basis vector of Λ is a nonzero lattice vector, and each step has unit length, so $\text{wt}(x) \geq |x| \geq \lambda_1$; hence $W(\Lambda) \geq \lceil \lambda_1 \rceil$. For the argmax tilings (oracle ids;

T -vectors are ζ_{12} -coefficient tuples from the catalogue; hexagonality $\omega T_1 \in \mathbb{Z}T_1 + \mathbb{Z}T_2$ verified exactly in each case, so $\lambda_1 = |T_1|$):

k	tiling	λ_1^2 (exact)	λ_1	$W \geq$	shell max wt	$W =$
1	4.6.12 (01_46c-3c-1)	$12 + 6\sqrt{3}$	4.732	5	5	5
2	02_346c-3c_41-1 ($p6m$)	$16 + 8\sqrt{3}$	5.464	6	6	6
3	03_346c-3a_4a_41-1 ($p6m$)	$21 + 12\sqrt{3}$	6.464	7	7	7
3	03_346c-3c_4h_4k-1 ($p6m$)	$28 + 10\sqrt{3}$	6.732	7	7	7

Each tiling’s minimal-vector shell has computed maximum weight equal to its lower bound, so the basis $(T_1, \omega T_1)$ realizes W exactly; combined with Theorem 1.1 the maxima over all $k \leq 3$ tilings are exactly 5, 6, 7, proving Theorem 1.2. (Consistent with the oracle measurement over all 92 tilings, which is used nowhere in the proofs.)

7 Consumption by the solver

The enumerator’s `poolConfig` has a proven mode (`poolSteps = 24k-1`, computationally dead) and a tuned mode (`poolSteps = 8` at $k = 3$), the latter below the proven hexagonal need of 10: *the existing $k = 3$ recertification (61/61) is not covered by this theorem*. Moreover the weight radius alone is not sufficient: the tuned pool also clips by Euclidean caps (`poolLmax` ≈ 8.12 at $k = 3$; the $|w|^2$ caps at 66), while the largest proven $k=3$ hexagonal shell is (40, 22) with $|u| \approx 8.84$, $|u|^2 \approx 78.1$ — a run with only `poolSteps` raised still loses shells (40, 22), (44, 16), (44, 14), (40, 16). The certified configuration must either set `poolLmax` ≥ 8.84 (hex; ≥ 5.99 square) and lift the $|w|^2$ caps accordingly, or bypass the pool with the explicit shell lists; the non-rigid branch additionally needs the generator-and-join pool $W(23) \cap \{|v| \leq \frac{2}{\sqrt{3}} \cdot 24.1\}$.

8 Adversarial verification

Three referee subagents attacked the source document independently (orbit/lattice theory; combinatorial geometry; computation audit), instructed to refute. *No fatal findings*. Survivals include: the census’s faithfulness to Lemma 3.1 through the non-symmorphic groups; the 10-species enumeration; the BFS weights, cross-checked against an independently derived closed form on all 4,140 shell elements and against a second referee’s reimplementation (CSV byte-identical); all five rigid shells and both attainment pins re-derived; the branch split’s exhaustiveness (union-of-pools consumption makes shape misclassification harmless); pool sizes exact. Gaps found and repaired in v2: the lattice-equality step of Lemma 3.5 (now the covering-radius argument); the propagation induction of Lemma 3.3 (was a clause, now written); the hypotheses of Lemma 3.4 (the brick counterexample); the Euclidean-cap requirement of Section 7 — the one finding that could have reproduced a real completeness hole downstream; citation and symbol fixes ($s_{\max}$ vs $a_{\max}$).

9 Status ledger

- **Proven** (cited): the Section 2 inputs. **Proven** (here): Lemmas 3.1 to 3.5, Theorem 1.2.
- **Computed** (exact, artifact): the Section 4 tables.

- **Slack, refinable, not claimed:** a snub-uniqueness lemma would give $R_1^{\text{hex}} = 5$; a triangle-region boundary-angle argument kills the $\{3^6, 3.12.12\}$ mixes and shaves $k = 2, 3$; the ST-6 quotient sweep should bring the non-rigid 23 down to ~ 8 .
- **OPEN for the completeness claim:** the re-run of the $k \leq 3$ enumeration at the proven configuration of Section 7. Until it exists, the $k = 3$ count 61 is oracle-matched, not certified.
- **Not claimed:** anything at $k \geq 4$ (the companion weight-ceiling document owns that regime); star families (dent-fill points change Lemma 3.1's graph substrate — work item TH-3).